Angel Fernando Borquez Guerrero

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Hermosillo, Sonora.

Education

Universidad de Sonora GPA: 91%
Bachelor of Science in Computer Science **August 2023 – Present**

Professional Experience

ACARUS — Supercomputing Area Supervisor: Dra. María del Carmen Heras Sánchez
Professional Intern **September 2025 – Present**

- Professional internship in the Supercomputing area at the University of Sonora.
- Part of a development team building a specialized forum for users of *Yuca*, the university HPC system.
- Implementing features such as user posts, direct messaging, and service request workflows.
- Supporting an active HPC user community by improving communication and technical support processes.

Research Experience

University of Guadalajara Supervisor: Dr. Rafael Morales Gamboa
Research Intern **June – July 2025**

- Generalized and validated an Entropic Associative Memory (EAM) model across multiple datasets through large-scale experimental evaluation and robustness analysis.
- Adapted and refactored the data pipeline to support class balancing, dynamic class selection, and scalable dataset handling.
- Adapted perceptual neural networks (autoencoder and classifier) to generate feature representations compatible with the EAM architecture.
- Designed and implemented a novelty detection experiment to evaluate the model's ability to reject unseen classes.
- Introduced experimental controls enabling partial memory registration and quantitative analysis of rejection behavior via no-response metrics.
- Refactored the system into a flexible experimental framework allowing controlled evaluations with a variable number of classes.

Competitions & Projects

NASA Space Apps Challenge 2024 Project: Voyager CXXV
Project Co-Lead **October 2024**

- Co-led the development of a geospatial simulation tool to evaluate the impact of urban green areas on temperature and air quality in Hermosillo, Mexico.
- Designed and implemented a system allowing users to define custom zones and simulate long-term environmental effects of urban reforestation.
- Integrated spatial data visualization through heat maps to support data-driven insights for urban planning scenarios.
- Coordinated technical tasks within the team, contributing to project planning, feature definition, and final integration.

TE AI CUP 2025 – International 3rd Place Project: AI Generated Machine Downtime Insights
UI Developer **January – May 2025**

- Designed and developed the UI/UX for an AI-based industrial downtime analysis platform, translating complex machine time-series data and unsupervised learning outputs into clear, actionable visual insights.
- Developed an accessible and user-centered interface for non-technical users, simplifying the exploration and interpretation of AI-generated insights without requiring prior knowledge of machine learning.
- Defined user flows, information hierarchy, and interaction patterns to improve navigation and facilitate the transition from high-level downtime trends to detailed analytical results.
- Collaborated with the data science and engineering teams to align the interface with the underlying analytical models and ensure that visual representations accurately reflected model outputs.

NASA Space Apps Challenge 2025 Project: Endurance CXXV
Project Co-Lead **October 2025**

- Co-led the development of an educational weather application designed for a children-oriented audience, focused on climate awareness and engagement.
- Developed both frontend and backend components of the application, contributing to core functionality and system integration.
- Designed and implemented the user interface with an emphasis on accessibility, usability, and age-appropriate interaction.

MICAI 2025 MeDA Challenge – National 1st Place **October 2025**

- Contributed to a 1st Place victory in a national academic competition focused on Medical Domain Adaptation using Self-Supervised Learning (SSL), leading to an invitation to publish in a special issue of the *Health Information Science and Systems* journal.
- Structured and standardized the dataset and training assets to establish a reproducible data pipeline for efficient model development and experimentation.
- Investigated approaches for cross-domain generalization and representation learning across heterogeneous medical imaging datasets.
- Developed visual and presentation assets used to communicate the project during the competition and dissemination stages.

SmartCity Hackathon – People's Choice Award Project: SmartTransit
Frontend Developer **March 2026**

- Developed the main frontend interface for an AI-based public transportation optimization system using demographic, geographic, and urban growth variables.
- Designed the UI structure and internal frontend logic to support route generation and optimization workflows.

HSIL Hackathon 2026 – 3rd Place Project: ShiftGuardian
Application Designer & AI Agent Architect **April 2026**

- Designed the application interface and AI agent architecture for a clinical decision support system supporting nurse handoffs and patient prioritization.
- Integrated a RAG-based LLM workflow concept to summarize clinical reports, identify critical tasks, and reduce cognitive load during shift transitions.

TE AI CUP 2026 – International 1st Place Project: Overwatch Alert Agent
Project Leader **January 2026 - June 2026**

- Led a 6-member interdisciplinary team in an international AI competition, guiding the technical development of an AI-driven system for industrial alert analysis that achieved 1st place.

- Developed an AI agent integrating Large Language Models (LLMs) for automated alert analysis and decision support.
- Developed an end-to-end data and machine learning pipeline involving data cleaning, text preprocessing, embedding generation, clustering, data validation, and predictive modeling.
- Developed a predictive modeling system to estimate the likelihood of future industrial alerts, providing a proactive component that complemented the analysis of historical patterns and enabled earlier identification of potential issues.
- Architected the project around clean, standardized, and scalable software structures, establishing conventions for modules, tools, configuration, state management, and component integration.

Technical Skills

Languages: Python, C++, Java, SQL, JavaScript, TypeScript
Frontend: HTML, CSS, SASS, Vue.js, NPM
Backend & Frameworks: Django
Systems & Infrastructure: Linux (Arch), HPC, Virtual Machines, Proxmox VE, VirtualBox
Tools: Git, GitHub, Jupyter, Google Colab, LaTeX